

Online Appendix to Cross-channel competition and complementarities in US retail

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O.1 Data processing

We chose the specialized offline retailers to analyze (Table 2) based on national store counts reported by Tables A1 through A3. In particular, we chose retailers that appear within the top three retailers in their respective categories for at least one year, excluding retailers that specialize in a narrow subcategory of their respective retailing categories (e.g., we do not include a large religious book retailer, Family Christian Book Store, in our analysis).

We identify offline stores in the book and office supplies categories by querying the Data Axle data for business locations with the following six-digit Standard Industrial Classification (SIC) codes: 594201 (“Book Stores”) for books and 594301 (“Office Supplies”) for office supplies. For electronics, we query for stores with the following four-digit SIC codes: 5731 (“Radio, Television, and Consumer Electronics Stores”), 5734 (“Computer and Computer Software Stores”), and 5946 (“Camera and Photographic Supply Stores”).

Table A1: Top book retailers by store count

Rank	2007		2008	
	Retailer	Count	Retailer	Count
1	Barnes and Noble	929	Barnes and Noble	929
2	Borders Books and Music	566	Borders Books and Music	566
3	Waldenbooks	383	Waldenbooks	383
4	Family Christian Book Store	269	Family Christian Book Store	269
5	Books-a-Million	177	Books-a-Million	177

Table A2: Top electronics retailers by store count

Rank	2007		2008	
	Retailer	Count	Retailer	Count
1	Radio Shack	5699	Radio Shack	5699
2	Best Buy	993	Best Buy	993
3	Circuit City	753	Circuit City	753
4	Ritz Camera Ctr	413	Ritz Camera Ctr	413
5	Compusa	235	Magnolia Home Theatre	235

Table A3: Top office supplies retailers by store count

Rank	2007		2008	
	Retailer	Count	Retailer	Count
1	Staples	1609	Staples	1609
2	Office Depot	1307	Office Depot	1307
3	Office Max	1068	Office Max	1068
4	Cartridge World	77	Corporate Express	77
5	Corporate Express	55	Indoff Inc	55

Our procedure for categorizing websites begins by identifying the most popular sites in the Comscore browsing data by visits and unique users. In particular, for each of the time periods (i) January to February 2007, (ii) November to December 2008, (iii) January to February 2017, and (iv) November to December 2018, we construct a list of the top 500 sites by the number of visits in the Comscore browsing data as well as a list of the top 500 sites by the number of unique visitors in these data. We then concatenate these lists and drop duplicated sites from the combined list. For each site, we manually determine in which of the aforementioned categories the site belongs. Some sites are not well described by any of these categories, and we do not place these sites in any category. Additionally, we do not place sites that are not in our list in any category.

De Los Santos et al. [2012] find that the Comscore data was largely representative of on-line buyers in the United States via a comparison of Comscore’s 2002 and 2004 datasets with data from the census and from a market research company. We arrive at a similar conclusion. Table A4 characterizes the representativeness of Comscore panelists for 2007

by comparing these panelists to demographic distributions reported in publications for the 2007 Current Population Survey (CPS) of the US census. The table suggests that the Comscore panelists are broadly representative of US internet users, with a few exceptions. We replicate the finding of De Los Santos et al. [2012] that Comscore over-samples Hispanic people relative to the share of Hispanic internet users reported by the CPS Computer and Internet Supplement. We additionally find that Comscore over-samples white people and under-samples Asian people. Note that we use the CPS for all households to obtain information on variables that are not available in the CPS Computer and Internet Use Supplement (e.g., household income and census region of residence).

One reason for which we focus on the 2007–2008 time period because the Comscore panel’s coverage of transactions is lower in 2017–2018, the other time period for which we have data. In 2017 and 2018, for example, there were respectively 92 and 101 million US members of Amazon Prime, Amazon’s premium subscription service.¹ Additionally, in an October 2017 survey, 92% of Prime members reported ordering from Amazon at least once a month, as did 61% of survey respondents who did not subscribe to Prime.² Yet only 18.3% of Comscore panelists made a transaction at Amazon in the 2017–2018 time period. Together, the facts that (i) most US consumers used Amazon in 2017–2018, (ii) Amazon reported only 81 million active accounts (Prime or non-Prime) in 2008,³ and (iii) the share of Comscore panelists using Amazon increased only from 14.3% in 2007–2008 to 18.3% in 2017–2018 suggest that the Comscore panel’s coverage of e-commerce transactions markedly decreased between our time periods.

¹Source: <https://www.digitalcommerce360.com/2019/07/11/82-of-us-households-have-a-amazon-prime-membership/>

²Source: <https://www.forbes.com/sites/louiscolumnbus/2018/03/04/10-charts-that-will-change-your-perspective-of-amazon-primes-growth/?sh=42b198b43fee>

³Source: <https://www.nytimes.com/2008/10/12/business/12giants.html>

Table A4: Representativeness of the Comscore Web Behavior Database for 2007

Variable	Comscore	CPS (All households)	CPS (Internet users)
Age: Under 24	0.02	0.06	0.06
Age: 25-34	0.15	0.16	0.18
Age: 35-44	0.27	0.19	0.23
Age: 45-54	0.28	0.21	0.24
Age: 55+	0.27	0.38	0.30
Hispanic	0.23	-	0.08
Race: White	0.94	-	0.84
Race: Black	0.05	-	0.09
Race: Asian	0.01	-	0.07
Race: Other	0.00	-	0.00
Household size: 1	0.06	0.28	-
Household size: 2	0.34	0.33	-
Household size: 3	0.24	0.16	-
Household size: 4	0.19	0.14	-
Household size: 5+	0.17	0.10	-
Census region: Northeast	0.19	0.18	-
Census region: North Central	0.22	0.22	-
Census region: South	0.39	0.37	-
Census region: West	0.20	0.22	-
Household income: Under 15k	0.13	0.13	-
Household income: 15-24k	0.08	0.12	-
Household income: 25-34k	0.10	0.11	-
Household income: 35-49k	0.15	0.14	-
Household income: 50-74k	0.22	0.18	-
Household income: 75-99k	0.14	0.11	-
Household income: 100k+	0.18	0.20	-
Broadband	0.87	-	0.82

Notes: This table compares the distribution of demographic variables among 2007 Comscore Web Behavior Database panelists with the distributions of these variables from the household-level 2007 Current Population Survey—which is labelled “CPS (All households)” in the table—and the 2007 Computer and Internet Use Supplement of the Current Population Survey, which is labelled “CPS (Internet users)” in the table. The table’s figures from the Computer and Internet Use Supplement describe the distribution of demographic variables within the population of householders that uses the internet.

O.2 Measures of rival effects and cross-channel complementarities

This section develops scale-free measures of offline stores' effects on online spending that facilitate comparisons across regressions. Let $\mathcal{J}^{\text{off}}$ ($\mathcal{J}^{\text{on}}$) denote the set of offline (online) retailers. We omit the retail category from our notation for simplicity, although we run our analysis separately for each category.

We first define a measure of the effect of an offline store j on the spending at an online store s . Consumer i 's expected spending at store s conditional on the regressors is

$$\mathbb{E}[y_{it}^s \mid x_{it}] = f(h(n_{it})'\alpha_s + z'_{it}\beta_s + w'_{it}\delta_s + \rho_{R_{it},s}^{\text{FE}} + \rho_{t,s}^{\text{FE}}) \quad (\text{A1})$$

We measure the effect of n_{ijt} on spending by the percentage change in expected spending when the number n_{ijt} of stores of offline retailer j is exogenously increased from $\bar{n}_j$ to $\bar{n}_j + 1$, where $\bar{x}$ denote the mean of a random variable x_{it} , while holding all other explanatory variables fixed at their mean values. That is, we define the relative effect of j on s as

$$\begin{aligned} \theta_{js} &= \frac{\mathbb{E}[y_i^s \mid \bar{n}_j + 1, \bar{n}_{-j}, \bar{z}, \bar{w}, \bar{\rho}_{R,s}^{\text{FE}}, \bar{\rho}_{t,s}^{\text{FE}}] - \mathbb{E}[y_i^s \mid \bar{n}_j, \bar{n}_{-j}, \bar{z}, \bar{w}, \bar{\rho}_{R,s}^{\text{FE}}, \bar{\rho}_{t,s}^{\text{FE}}]}{\mathbb{E}[y_i^s \mid \bar{n}_j, \bar{n}_{-j}, \bar{z}, \bar{w}, \bar{\rho}_{R,s}^{\text{FE}}, \bar{\rho}_{t,s}^{\text{FE}}]} \\ &= \frac{f(h(\bar{n}_j + 1, \bar{n}_{-j})'\alpha_s + \bar{z}'\beta_s + \bar{w}'\delta_s + \bar{\rho}_{R,s}^{\text{FE}} + \bar{\rho}_{t,s}^{\text{FE}})}{f(h(\bar{n}_j, \bar{n}_{-j})'\alpha_s + \bar{z}'\beta_s + \bar{w}'\delta_s + \bar{\rho}_{R,s}^{\text{FE}} + \bar{\rho}_{t,s}^{\text{FE}})} - 1 \end{aligned} \quad (\text{A2})$$

We estimate θ_{js} by substituting our estimates of unknown parameters into (A2).

Using the store-pair-specific effects defined by (A2), we define the measures of rival and own effects. First, define the *store-specific average rival effect* as

$$\theta_s^{\text{rival}} = \sum_{j \in \mathcal{J}^{\text{off}} \setminus \{s\}} w_{js}^{\text{off}} \theta_{js} \quad (\text{A3})$$

where $\{w_{js}^{\text{off}}\}_j$ are weights for offline retailers j satisfying $\sum_{j \in \mathcal{J}^{\text{off}}} w_{js}^{\text{off}} = 1$ and $w_{ss}^{\text{off}} = 0$. We set w_{js}^{off} proportional to retailer j 's number of stores in the analysis period.

The rival effect defined in (A3) includes all effects of rival offline stores on online spending. Therefore, a positive value for this measure would indicate that a showrooming effect outweighs the competitive effect as long as the former effect is positive and the latter is negative. The converse of this statement is true for a negative rival effect.

We take θ_{ss} as a measure of *store-specific own effect*. This measure will be positive (negative) if cross-channel complementarities are larger (smaller) than cannibalization net of showrooming effects, assuming that the former are positive and the latter are negative.

We also interpret averages of our store-specific average rival and own effects for each retailing category in each time period. These averages are defined as

$$\bar{\theta}^{\text{rival}} = \sum_{s \in \mathcal{J}^{\text{on}} \cap \mathcal{J}^{\text{off}}} w_s^{\text{on}} \theta_s^{\text{rival}} \quad (\text{A4})$$

$$\bar{\theta}^{\text{own}} = \sum_{s \in \mathcal{J}^{\text{on}} \cap \mathcal{J}^{\text{off}}} w_s^{\text{on}} \theta_{ss} \quad (\text{A5})$$

for weights $\{w_s^{\text{on}}\}_s$ on stores s that sum to one across s . In practice, we set w_s^{on} proportional to the mean of y_i^s . Note that each of the average measures defined above is taken over multichannel retailers $s \in \mathcal{J}^{\text{on}} \cap \mathcal{J}^{\text{off}}$. We also compute an average of rival effects over all online retailers, including Amazon:

$$\bar{\theta}_{\text{incl}}^{\text{rival}} = \sum_{s \in \mathcal{J}^{\text{on}}} w_s^{\text{on}} \theta_s^{\text{rival}}.$$

Alongside these two averages, we report Amazon's rival measure, θ_s^{rival} for $s = \text{Amazon}$, separately.

0.3 Store-level effects on the probability of positive spending

Table A5 reports the store-level rival and own effects on the probability of positive online spending. They exhibit patterns similar to the store-level rival and own effects on online expenditures, reported in Tables 8 through 11. Specifically, the own effects are positive, the rival effects are mostly negative, and the rival effects on Amazon tend to be less negative (or even positive).

Table A5: Store-level rival and own effects on the probability of positive spending

(a) Cross-category retailers

	amazon (1)	costco.com (2)	target.com (3)	walmart.com (4)	Average (5)
Rival	-0.001 (0.002)	-0.017 (0.009)	-0.016 (0.006)	-0.030 (0.004)	-0.024 (0.003)
Own		0.402 (0.061)	0.023 (0.008)	0.044 (0.005)	0.056 (0.005)

(b) Bookstores

	amazon (1)	barnesandnoble.com (2)	booksamillion.com (3)	Average (4)
Rival	0.006 (0.002)	-0.018 (0.004)	-0.031 (0.013)	-0.019 (0.004)
Own		0.118 (0.015)	0.743 (0.195)	0.160 (0.019)

(c) Electronics

	amazon (1)	apple.com (2)	bestbuy.com (3)	circuitcity.com (4)	radioshack.com (5)	Average (6)
Rival	0.003 (0.002)	0.007 (0.004)	-0.011 (0.004)	-0.015 (0.004)	0.005 (0.015)	-0.007 (0.002)
Own		0.028 (0.044)	0.084 (0.028)	0.155 (0.036)	0.002 (0.020)	0.087 (0.019)

(d) Office supplies

	amazon (1)	officedepot.com (2)	officemax.com (3)	staples.com (4)	Average (5)
Rival	0.018 (0.016)	-0.013 (0.006)	-0.021 (0.012)	-0.030 (0.005)	-0.023 (0.004)
Own		0.161 (0.021)	0.159 (0.057)	0.094 (0.011)	0.124 (0.011)

Note: Each panel reports rival and own effects on the probability of positive spending (percentage changes in the probability), computed from Poisson regressions of positive-spending indicators on store counts with the same controls as the main specification. The “Average” column reports the weighted average of the effects across the category’s multichannel online stores (i.e., excluding Amazon), with weights proportional to each store’s share of consumers with positive spending. Standard errors appear in parentheses.

O.4 Regression results and store-level effects with disc instruments

Table A6 summarizes the first-stage results of the regressions using the disc instrumental variables (IVs) described in Section 4.2, and Tables A7 through A10 present the estimates of the store-level regression coefficients and rival/own effects for each category. The first-stage F statistics, obtained by partialling out other controls, indicate that the disc IVs are strong.

Table A7 presents the estimates of the regression coefficients and effects for cross-category retailers. The rival effects are mostly negative, although estimates are less statistically significant than our main estimates. Also, among own-store coefficients, only Costco's is positive and statistically significant. When aggregated to the level of online stores, the rival effect is negative for all stores, and statistically significant for Amazon and Target. The own effect is positive and marginally significant for Costco, and imprecisely estimated for the other multichannel stores.

Table A8 presents the results for bookstores. Although the regression estimates are again less precise than our main estimates, the rival-store coefficients tend to be negative, and the own-store coefficient for Barnes & Noble is positive and statistically significant. For Barnes & Noble, the rival effect is negative and the own effect is positive, both in a statistically significant manner.

Tables A9 and A10 show the results for electronics and office supplies, respectively. For these categories, the regression estimates are much noisier, as are the store-level effects.

O.5 Robustness to alternative specifications

In this section, we evaluate robustness of our main results shown in Section 5 to alternative specifications. Specifically, we show that the qualitative results reported in Tables 6 and 7 remain largely unchanged if we change functional forms (from Poisson to linear), the definition of local markets (from 20km distance bands to 10km or 50km distance bands),

Table A6: First-stage results for the disc instruments approach

Store count	R^2	F
<i>Booksellers</i>		
Barnes	0.18	588.30
Books-a-Million	0.08	236.48
Borders	0.13	426.33
Waldenbooks	0.06	165.34
<i>Electronics retailers</i>		
Apple	0.21	722.12
Best Buy	0.09	270.54
Circuit City	0.08	244.37
Radio Shack	0.12	387.28
<i>Office-supply retailers</i>		
Office Depot	0.08	226.70
Office Max	0.09	278.26
Staples	0.10	298.89
<i>Cross-category retailers</i>		
Costco	0.10	321.23
Target	0.06	169.25
Walmart	0.13	401.98
<i>Total store count, by category</i>		
Cross-category	0.08	223.91
Books	0.22	766.42
Electronics	0.06	178.65
Office	0.13	397.19

Notes: each entry summarizes the first stage for one endogenous store-count regressor in the disc-instrument version of the spending model (Section 4.2), with one row per store. The endogenous variable is the number of the indicated chain’s offline stores within 20 km of the consumer; the rows in the final panel instead use the total number of a category’s stores within 20 km, the regressor in that category’s overall-spending specification. Following the control-function procedure of Section 4.2, each store count is regressed on the disc instruments of Cao et al. [2024] together with all consumer characteristics, web-browsing controls, and fixed effects described in the main text. The excluded instruments are the baseline disc IVs, constructed for race, Hispanic origin, presence of children, household size, and income and each interacted with the consumer’s own group, together with their population-weighted counterparts. We report diagnostics for these instruments after the controls are partialled out of both the store count and the instruments: R^2 is the partial R^2 attributable to the disc instruments alone, and F is the joint F -statistic for the excluded instruments in the partialled-out regression. Larger values indicate stronger instruments; all values here lie well above conventional weak-instrument thresholds. A separate first stage is estimated for each store-count variable. Because each store’s first stage does not depend on the spending category, a chain that enters more than one category’s spending regressions has identical R^2 and F throughout and is reported once, with stores grouped by retailer type. The estimation sample is all 2007–2008 consumers with non-missing disc instruments.

Table A7: Store-specific cross-category spending (disc instruments)

(a) Coefficients

	Spending			
	amazon	costco.com	target.com	walmart.com
	(1)	(2)	(3)	(4)
N. Stores: Costco	-0.052 (0.125)	1.587 (0.633)	-0.138 (0.231)	0.097 (0.199)
N. Stores: Target	0.142 (0.146)	-0.374 (0.680)	-0.180 (0.266)	-0.322 (0.231)
N. Stores: Walmart	-0.426 (0.093)	-0.295 (0.363)	-0.614 (0.174)	0.009 (0.150)
Mean dep. var.	14.23	2.64	3.21	5.51
Observations	136,997	137,378	137,298	137,232

(b) Rival effects and own effects

	amazon	costco.com	target.com	walmart.com	Average
	(1)	(2)	(3)	(4)	(5)
Rival	-0.026 (0.013)	-0.059 (0.052)	-0.089 (0.039)	-0.029 (0.029)	-0.053 (0.021)
Own		0.826 (0.438)	-0.034 (0.049)	0.002 (0.027)	0.183 (0.104)

Note: The “Average” column reports the weighted average of the effects across the category’s multi-channel online stores (i.e., excluding Amazon), with weights proportional to mean consumer spending at each online store. Standard errors appear in parentheses.

and region fixed effects (from census regions to states).

Tables A11 through A14 present the results of overall regressions with the aforementioned specification changes. All the estimates show the same sign as the main results in Table 6: the coefficients on the number of stores are negative for cross-category retailers whereas they are positive for specialized retailers. Although the estimates are occasionally imprecise, they are robustly precise for books.

Tables A15 through A18 display category-level rival and own effects under alternative regression specifications. The results are again similar to the main results. Specifically,

Table A8: Store-specific book spending (disc instruments)

(a) Coefficients

	Spending		
	amazon	barnesandnoble.com	booksamillion.com
	(1)	(2)	(3)
N. Stores: Barnes	0.111 (0.254)	1.231 (0.531)	-1.025 (1.970)
N. Stores: Books-a-Million	0.179 (0.210)	-0.030 (0.450)	-0.069 (1.868)
N. Stores: Borders	-0.573 (0.223)	-0.358 (0.451)	1.846 (1.882)
N. Stores: Waldenbooks	-0.205 (0.228)	-0.987 (0.497)	0.689 (1.648)
Mean dep. var.	5.57	0.88	0.06
Observations	137,164	137,343	137,390

(b) Rival effects and own effects

	amazon	barnesandnoble.com	booksamillion.com	Average
	(1)	(2)	(3)	(4)
Rival	-0.019 (0.013)	-0.062 (0.029)	0.100 (0.179)	-0.051 (0.029)
Own		0.331 (0.164)	-0.039 (1.051)	0.307 (0.168)

Note: The “Average” column reports the weighted average of the effects across the category’s multi-channel online stores (i.e., excluding Amazon), with weights proportional to mean consumer spending at each online store. Standard errors appear in parentheses.

in all specifications, the own effects are positive and the rival effects are negative among multichannel retailers, and the rival effects become less negative (and positive for bookstores in particular) when Amazon is included.

Table A9: Store-specific electronics spending (disc instruments)

(a) Coefficients

	Spending				
	amazon	apple.com	bestbuy.com	circuitcity.com	radioshack.com
	(1)	(2)	(3)	(4)	(5)
N. Stores: Apple	0.391 (0.393)	0.122 (0.675)	-0.230 (0.510)	0.638 (0.607)	0.062 (1.144)
N. Stores: Best Buy	-0.890 (0.862)	-0.881 (1.304)	-0.513 (1.051)	-1.241 (1.261)	-3.663 (2.177)
N. Stores: Circuit City	0.569 (0.855)	0.993 (1.689)	-1.779 (1.125)	1.347 (1.394)	8.318 (3.438)
N. Stores: Radio Shack	0.375 (0.872)	0.687 (1.536)	1.205 (1.089)	-1.017 (1.159)	-0.638 (3.021)
Mean dep. var.	3.26	2.44	2.34	2.21	0.08
Observations	137,343	137,375	137,369	137,372	137,391

(b) Rival effects and own effects

	amazon	apple.com	bestbuy.com	circuitcity.com	radioshack.com	Average
	(1)	(2)	(3)	(4)	(5)	(6)
Rival	0.011 (0.022)	0.032 (0.054)	-0.023 (0.021)	-0.044 (0.030)	1.376 (1.605)	0.004 (0.028)
Own		0.058 (0.327)	-0.122 (0.235)	0.461 (0.573)	-0.045 (0.210)	0.123 (0.225)

Note: The “Average” column reports the weighted average of the effects across the category’s multi-channel online stores (i.e., excluding Amazon), with weights proportional to mean consumer spending at each online store. Standard errors appear in parentheses.

Table A10: Store-specific office supplies spending (disc instruments)

(a) Coefficients

	Spending			
	amazon	officedepot.com	officemax.com	staples.com
	(1)	(2)	(3)	(4)
N. Stores: Office Depot	2.057 (2.180)	1.941 (1.141)	2.682 (2.368)	-0.548 (1.029)
N. Stores: Office Max	-1.217 (0.930)	0.509 (0.437)	-1.064 (1.180)	0.409 (0.503)
N. Stores: Staples	-0.182 (1.013)	-0.344 (0.574)	1.211 (1.103)	0.328 (0.419)
Mean dep. var.	0.07	3.62	0.34	4.57
Observations	137,391	137,378	137,390	137,370

(b) Rival effects and own effects

	amazon	officedepot.com	officemax.com	staples.com	Average
	(1)	(2)	(3)	(4)	(5)
Rival	0.008 (0.128)	-0.014 (0.048)	0.191 (0.212)	-0.029 (0.035)	-0.014 (0.029)
Own		0.470 (0.333)	-0.242 (0.233)	0.050 (0.065)	0.217 (0.146)

Note: The “Average” column reports the weighted average of the effects across the category’s multi-channel online stores (i.e., excluding Amazon), with weights proportional to mean consumer spending at each online store. Standard errors appear in parentheses.

Table A11: Overall spending regressions (linear)

	Cross-category retailers (1)	Bookstores (2)	Electronics (3)	Office supplies (4)
N. Stores: Total	-10.493 (2.863)	0.914 (0.209)	2.792 (1.203)	1.603 (0.891)
Mean dep. var.	187.35	9.14	47.37	12.91
Observations	145,345	146,506	146,404	146,765

Table A12: Overall spending regressions (10km)

	Cross-category retailers (1)	Bookstores (2)	Electronics (3)	Office supplies (4)
N. Stores: Total	-0.030 (0.014)	0.117 (0.020)	0.038 (0.021)	0.158 (0.055)
Mean dep. var.	187.35	9.14	47.37	12.91
Observations	145,345	146,506	146,404	146,765

Table A13: Overall spending regressions (50km)

	Cross-category retailers (1)	Bookstores (2)	Electronics (3)	Office supplies (4)
N. Stores: Total	-0.100 (0.024)	0.190 (0.036)	0.116 (0.044)	0.041 (0.107)
Mean dep. var.	187.35	9.14	47.37	12.91
Observations	145,345	146,506	146,404	146,765

Table A14: Overall spending regressions (state fixed effects)

	Cross-category retailers (1)	Bookstores (2)	Electronics (3)	Office supplies (4)
N. Stores: Total	-0.018 (0.017)	0.097 (0.025)	0.060 (0.027)	0.178 (0.072)
Mean dep. var.	187.35	9.14	47.37	12.91
Observations	145,345	146,506	146,404	146,765

Note: Tables A11 through A14 present estimated coefficients from regressions of the overall spending on offline store counts, using a linear specification (A11), 10km distance bands (A12), 50km distance bands (A13), or state fixed effects (A14) instead of the main specification in Table 6. The “Mean dep. var” row presents the averages of the dependent variable (expenditures in dollars). Heteroskedasticity-robust standard errors in parentheses.

Table A15: Category-level rival and own effects on expenditures (linear)

	Cross-category retailers (1)	Bookstores (2)	Electronics (3)	Office supplies (4)
Rival (multichannel)	-0.016 (0.008)	-0.011 (0.023)	-0.006 (0.057)	-0.005 (0.017)
Rival (Amazon)	-0.006 (0.003)	0.008 (0.009)	0.006 (0.030)	0.033 (0.121)
Rival (all retailers)	-0.010 (0.004)	0.005 (0.008)	-0.003 (0.040)	-0.005 (0.017)
Own	0.112 (0.044)	0.176 (0.419)	0.045 (0.343)	0.116 (0.105)

Note: Each column presents the category-level average rival effects and own effects as described in Section 4.3. “Rival (multichannel)” and “Rival (all retailers)” average the rival effects across the multichannel retailers’ online stores and across all online stores (including Amazon), respectively, weighting each store in proportion to mean consumer spending; “Rival (Amazon)” is the rival effect on Amazon’s online sales. “Own” averages the own effects across the multichannel retailers’ online stores.

Table A16: Category-level rival and own effects on expenditures (10km)

	Cross-category retailers (1)	Bookstores (2)	Electronics (3)	Office supplies (4)
Rival (multichannel)	-0.034 (0.010)	-0.024 (0.007)	-0.006 (0.006)	-0.006 (0.012)
Rival (Amazon)	-0.014 (0.005)	0.012 (0.004)	0.012 (0.005)	-0.014 (0.062)
Rival (all retailers)	-0.023 (0.005)	0.007 (0.003)	-0.001 (0.004)	-0.006 (0.012)
Own	0.135 (0.033)	0.244 (0.047)	0.075 (0.051)	0.280 (0.043)

Note: Each column presents the category-level average rival effects and own effects, computed using store-specific regressions with 10km distance bands to define local markets. “Rival (multichannel)” and “Rival (all retailers)” average the rival effects across the multichannel retailers’ online stores and across all online stores (including Amazon), respectively, weighting each store in proportion to mean consumer spending; “Rival (Amazon)” is the rival effect on Amazon’s online sales. “Own” averages the own effects across the multichannel retailers’ online stores.

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Table A17: Category-level rival and own effects on expenditures (50km)

	Cross-category retailers	Bookstores	Electronics	Office supplies
	(1)	(2)	(3)	(4)
Rival (multichannel)	-0.005 (0.002)	-0.002 (0.003)	-0.003 (0.002)	-0.004 (0.002)
Rival (Amazon)	0.002 (0.001)	0.004 (0.001)	0.006 (0.002)	0.011 (0.009)
Rival (all retailers)	-0.001 (0.001)	0.003 (0.001)	-0.000 (0.001)	-0.004 (0.002)
Own	0.046 (0.008)	0.068 (0.013)	0.020 (0.017)	0.061 (0.008)

Note: Each column presents the category-level average rival effects and own effects, computed using store-specific regressions with 50km distance bands to define local markets. “Rival (multichannel)” and “Rival (all retailers)” average the rival effects across the multichannel retailers’ online stores and across all online stores (including Amazon), respectively, weighting each store in proportion to mean consumer spending; “Rival (Amazon)” is the rival effect on Amazon’s online sales. “Own” averages the own effects across the multichannel retailers’ online stores.

Table A18: Category-level rival and own effects on expenditures (state fixed effects)

	Cross-category retailers	Bookstores	Electronics	Office supplies
	(1)	(2)	(3)	(4)
Rival (multichannel)	-0.014 (0.006)	-0.014 (0.006)	-0.007 (0.004)	-0.001 (0.007)
Rival (Amazon)	-0.005 (0.003)	0.007 (0.003)	0.006 (0.004)	0.022 (0.027)
Rival (all retailers)	-0.009 (0.003)	0.004 (0.003)	-0.003 (0.003)	-0.001 (0.007)
Own	0.117 (0.023)	0.175 (0.031)	0.052 (0.034)	0.127 (0.022)

Note: Each column presents the category-level average rival effect and own effects, computed using store-specific regressions with state fixed effects. “Rival (multichannel)” and “Rival (all retailers)” average the rival effects across the multichannel retailers’ online stores and across all online stores (including Amazon), respectively, weighting each store in proportion to mean consumer spending; “Rival (Amazon)” is the rival effect on Amazon’s online sales. “Own” averages the own effects across the multichannel retailers’ online stores.

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